B. Almost Ternary Matrix

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: bitmasks, constructive algorithms, matrices | https://codeforces.com/problemset/problem/1699/B

You are given two **even** integers n and m. Your task is to find **any** binary matrix a with n rows and m columns where every cell (i,j) has **exactly** two neighbours with a different value than $a_{i,j}$.

Two cells in the matrix are considered neighbours if and only if they share a side. More formally, the neighbours of cell (x,y) are: $(x-1,y)$, $(x,y+1)$, $(x+1,y)$ and $(x,y-1)$.

It can be proven that under the given constraints, an answer always exists.

Input

Each test contains multiple test cases. The first line of input contains a single integer t ($1 \le t \le 100$) — the number of test cases. The following lines contain the descriptions of the test cases.

The only line of each test case contains two **even** integers n and m ($2 \le n,m \le 50$) — the height and width of the binary matrix, respectively.

Output

For each test case, print n lines, each of which contains m numbers, equal to 0 or 1 — any binary matrix which satisfies the constraints described in the statement.

It can be proven that under the given constraints, an answer always exists.

Example

Input
3 2 4 2 2 4 4

Output
1 0 0 1 0 1 1 0 1 0 0 1 1 0 1 0 0 0 1 1 1 1 0 0 0 1 0 1

Note

White means 0, black means 1.

The binary matrix from the first test case	The binary matrix from the second test case	The binary matrix from the third test case